Zhaoxiang Feng

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Seeking ML Systems, LLM Inference, and ML Research Internships | Available May–August 2027

Education

The University of Texas at Austin — Austin, TX
Master of Science in Computer Science — *Aug. 2026 – Present*

University of California San Diego — San Diego, CA
B.S. in Data Science; B.S. in Probability & Statistics; GPA: 3.84/4.00 — *Sep. 2022 – Mar. 2026*

Research Experience

Hao AI Lab, UC San Diego — San Diego, CA
Research Assistant, Advisor: Prof. Hao Zhang — *Sep. 2025 – Present*

- Built JetSpec's standalone **LLM inference engine**: **paged KV cache**, continuous batching, a paged tree-attention **Triton kernel**, and **CUDA-graph** verification, serving **∼1,000 tokens/s** on one NVIDIA B200
- Contributed to **JetSpec**, drafting a full speculative tree in one forward pass: up to **9.64**× end-to-end speedup over autoregressive decoding on Qwen3-8B; second author, under review at NeurIPS 2026
- Led **JetSpec++**, conditioning an attention-layer causal correction on each token's full branch history; accepted **0.89 more tokens per drafting round** than DeepSeek's DSpark at matched parameter count, widening to **1.67** at draft depths unseen in training

Q-Lab, UC San Diego — San Diego, CA
Research Assistant, Advisor: Prof. Lianhui Qin — *Jan. 2025 – Jun. 2026*

- Contributed to **ArcMemo**'s typed, composable memory for reusable reasoning concepts and its reasoning-guided concept selection; **Runner-Up, ARC Prize 2025 Paper Awards**
- Built a **concept-to-puzzle pipeline** that generates executable puzzle-generator code from concepts and filters its outputs for well-formedness and determinism

Wang Lab, UC San Diego — San Diego, CA
Student Researcher, Advisor: Prof. Wei Wang — *Jun. 2025 – Jun. 2026*

- Developing **TrustPPI**, a reliability signal from embedding-perturbation stability for protein interaction models; **0.66–0.83 AUROC** error detection across three architectures; manuscript in preparation

Projects

DFlash on Google Cloud TPU | *JAX, vLLM tpu-inference, TPU v4/v5p* — Code | Google Developers Blog

- Built DFlash block-diffusion speculative decoding on TPU in JAX with the Google Cloud TPU team: a standalone draft-verify loop with dual KV cache, integrated into vLLM's serving path and speculative decoding manager
- Upstreamed a **2.8K-line PR** into vLLM tpu-inference (#1868, merged): DFlash draft model, concat-attention kernel, JAX and torchax proposers, and unit tests
- Reached **3.13× average speedup** over nine benchmarks on TPU v5p (up to 5.72× on MATH-500) and 2.29× vs. 1.30× for EAGLE-3, after fixing a sequence-length alignment bug that depressed acceptance
- Showed verification cost stays flat in draft block size through **K=1024** on TPU, and built a GPU training suite for wider draft blocks (K=32–128)

LLM Pre-training System | *PyTorch, DDP, ZeRO-1, FlashAttention-2* — Sep. 2025 – Dec. 2025

- Built the training stack for a **1.8B-parameter** model on **8 NVIDIA B200 GPUs**, using BF16 and torch.compile; achieved **160K tokens/s** and **36.7% model FLOPs utilization**
- Implemented **SFT and DPO** post-training on Alpaca and HH-RLHF; supervised fine-tuning improved OpenBookQA accuracy by **2.6 percentage points** over the base model

Technical Skills

Languages: Python, R, SQL, Java, JavaScript
ML Systems: PyTorch, JAX, Triton, vLLM, Hugging Face Transformers, DDP, ZeRO-1, CUDA graphs
Tools: Git, Linux, Docker